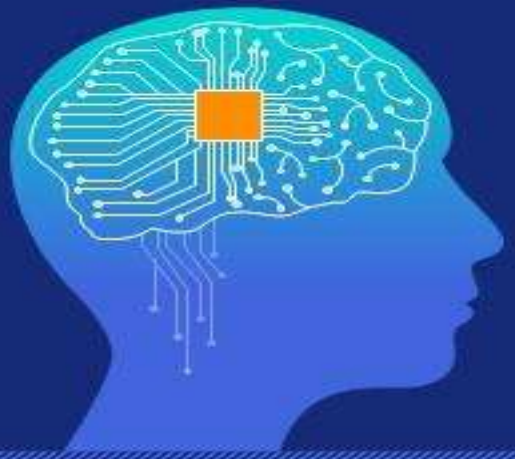
Deep Learning



Sayed Mostafa Modaghegh

What is Deep Learning?



Deep Learning is a subfield of machine learning that is concerned with algorithms inspired by the brain's structure and functions known as artificial neural networks.

A computer model can be taught using Deep Learning to run classification actions using pictures, texts or sounds as input.

What is Deep Learning?



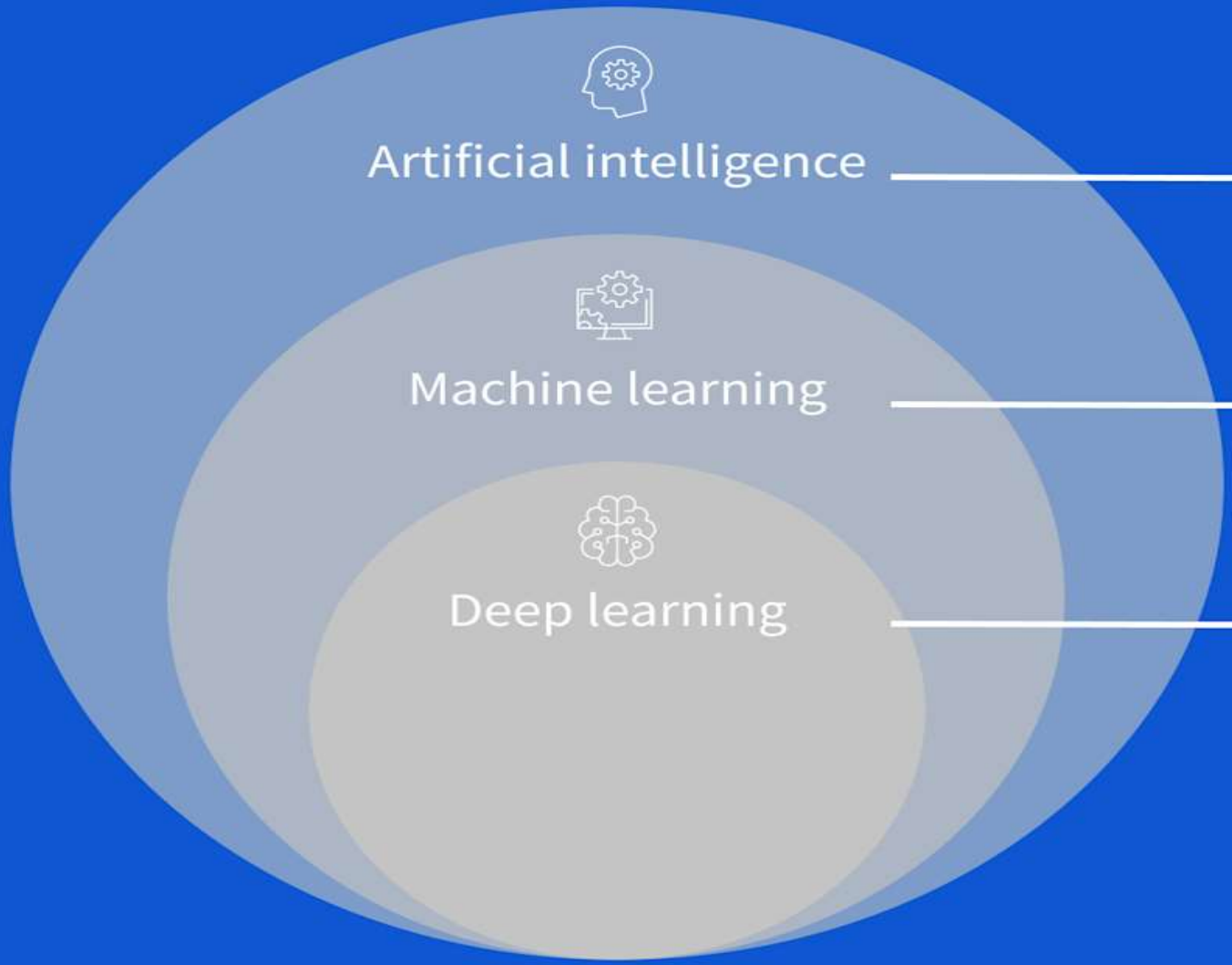


Car
Not Car

Input

Feature Extraction + Classification

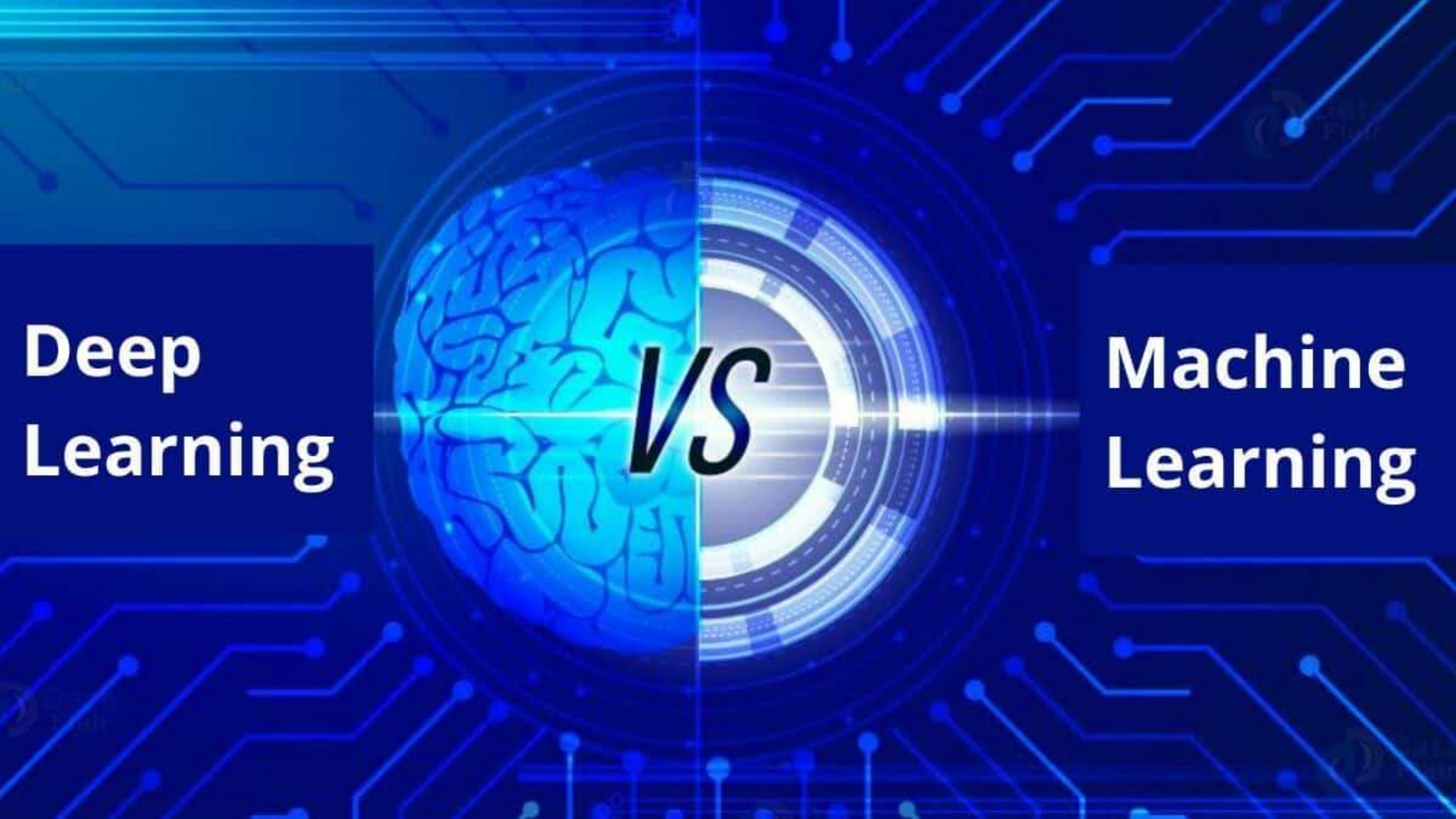
Output



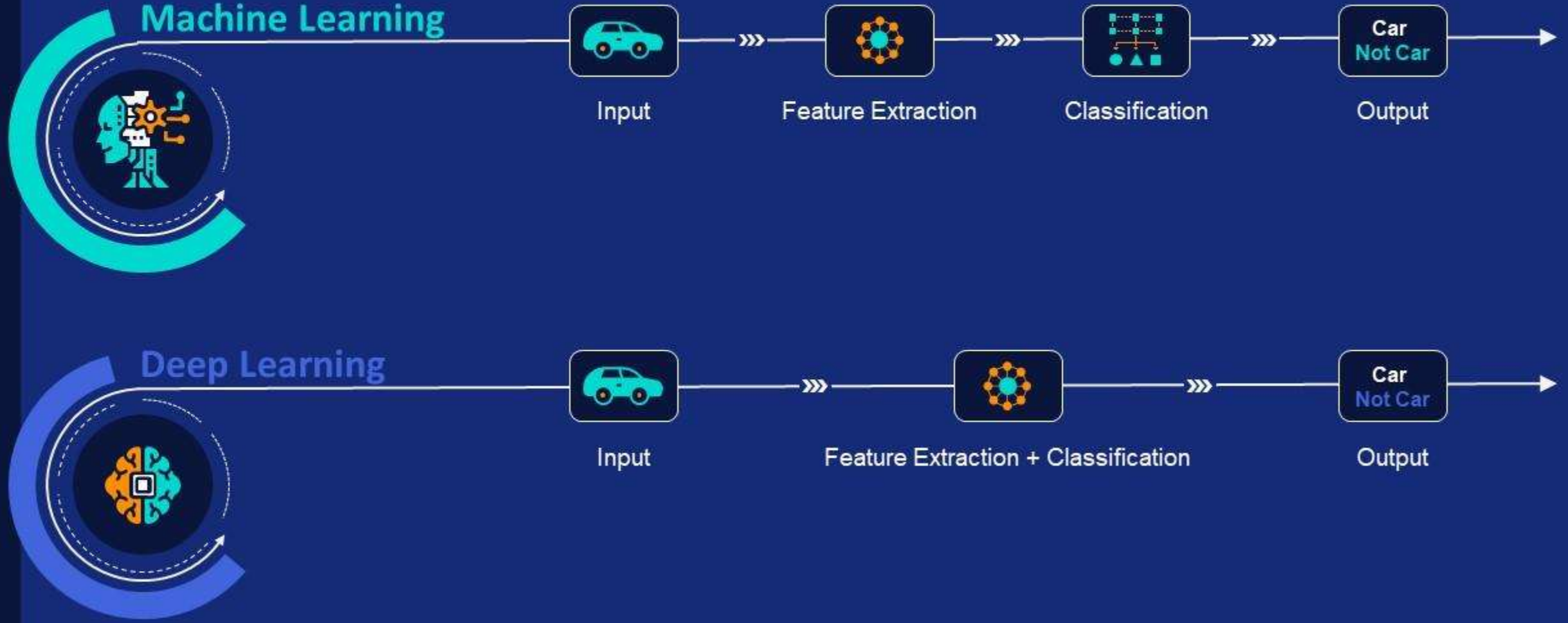
Artificial intelligence uses computer science and data to enable problem solving in machines

Machine learning refers to the study of computer systems that learn and adapt automatically from experience, without being explicitly programmed

Deep learning is a machine learning technique that layers algorithms and computing units—or neurons—into artificial neural networks that mimic the human brain


Deep Learning
VS
Machine Learning

Difference between Machine Learning and Deep Learning



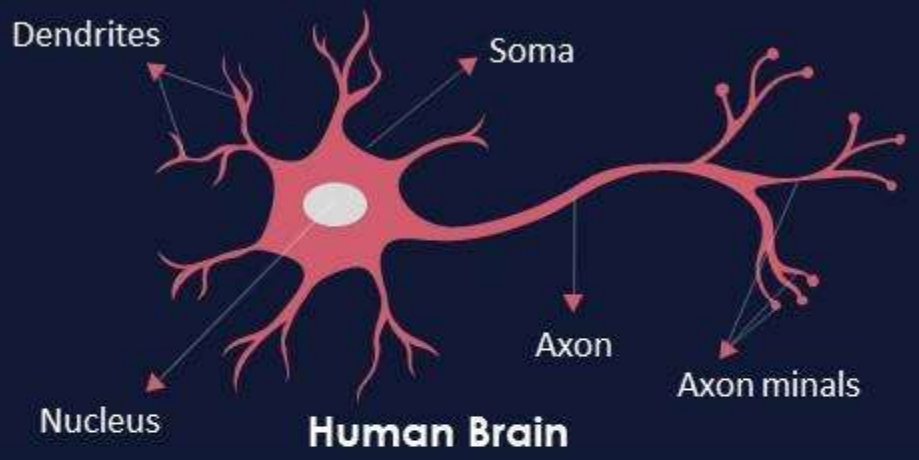


VS

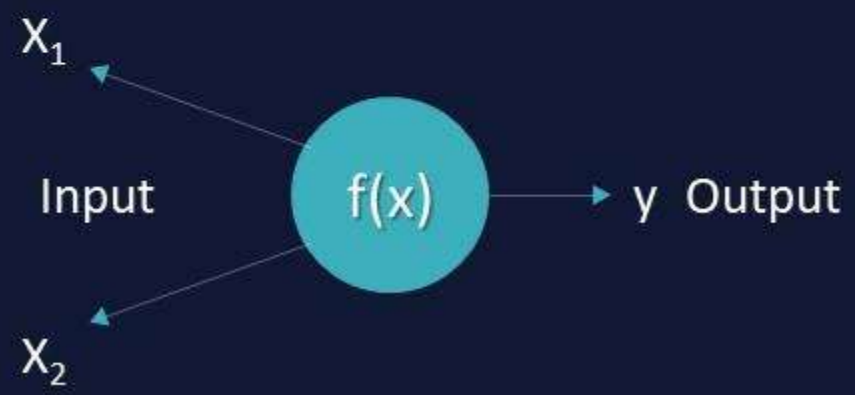


DEEP LEARNING		MACHINE LEARNING
It is a subset of Machine Learning.	HIERARCHICAL CLASSIFICATION	It is a subpart of Artificial Intelligence.
It can work with a broad range of data, including structured and unstructured data.	DATA VERSATILITY	It can mainly work with structured data.
Deep Learning uses millions of data points.	DATA REQUIRED	Machine Learning uses thousands of data points.
Deep Learning has a longer training time.	TRAINING TIME	Machine Learning has a shorter training time.
The setting-up time for Deep Learning systems may be longer.	SETUP TIME	The setup time for ML systems is shorter.
Deep Learning requires powerful resources to function.	COMPUTING RESOURCES	Machine Learning can work on a standard computer system.
Deep Learning can create features on its own.	FEATURE ENGINEERING	Machine Learning relies on users to create features.

Neurons in a Neural Network

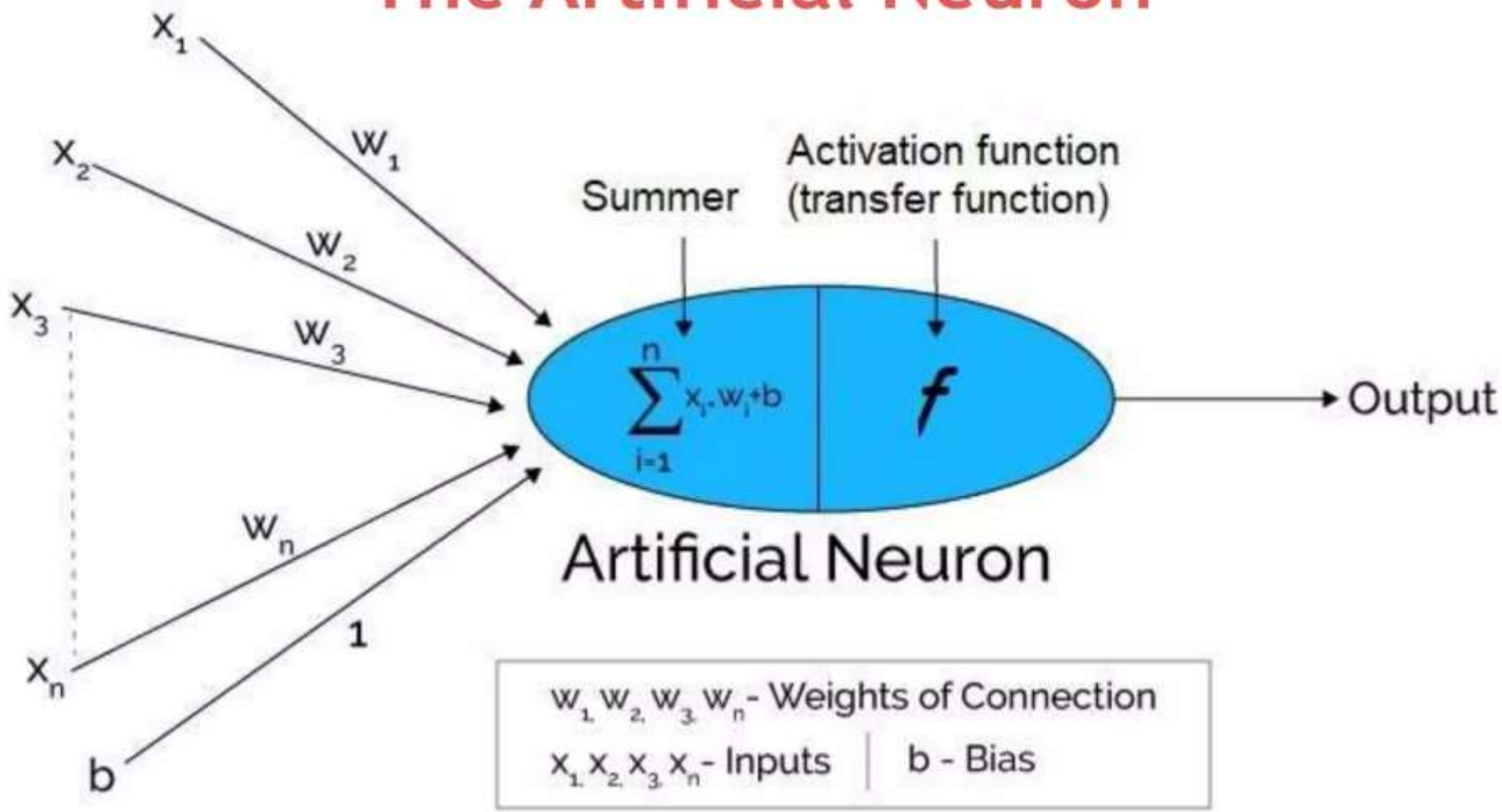


Biological Neuron



Artificial Neuron

The Artificial Neuron



w_1, w_2, w_3, w_n - Weights of Connection

x_1, x_2, x_3, x_n - Inputs | b - Bias

$$Output = f\left(\sum_{i=0}^{n} w_i x_i + b\right)$$

Activation Functions in Neural Networks

Activation Function in a Neural Network

What is an Activation Function?

The activation function is used to calculate a weighted total and then add bias to it to determine whether a neuron should be activated or not.

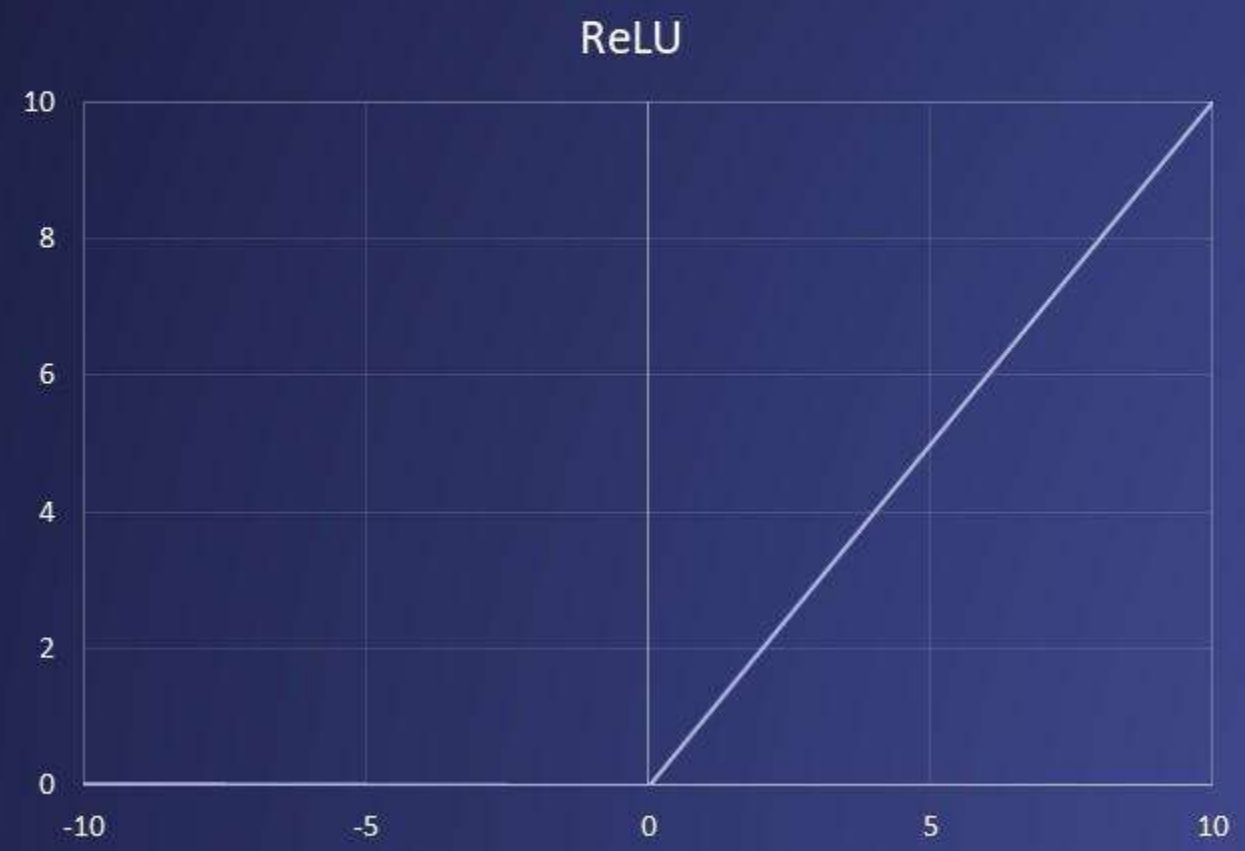
Why do we need activation functions?

Without an activation function, a neural network is just a linear regression model. The activation function transforms the input in a non-linear way, allowing it to learn and accomplish more complex tasks.



ReLU (Rectified Linear Unit) activation function

This slide demonstrates the rectified linear unit activation function, which is the most utilized function in the current world and is both derivative and monotonic.

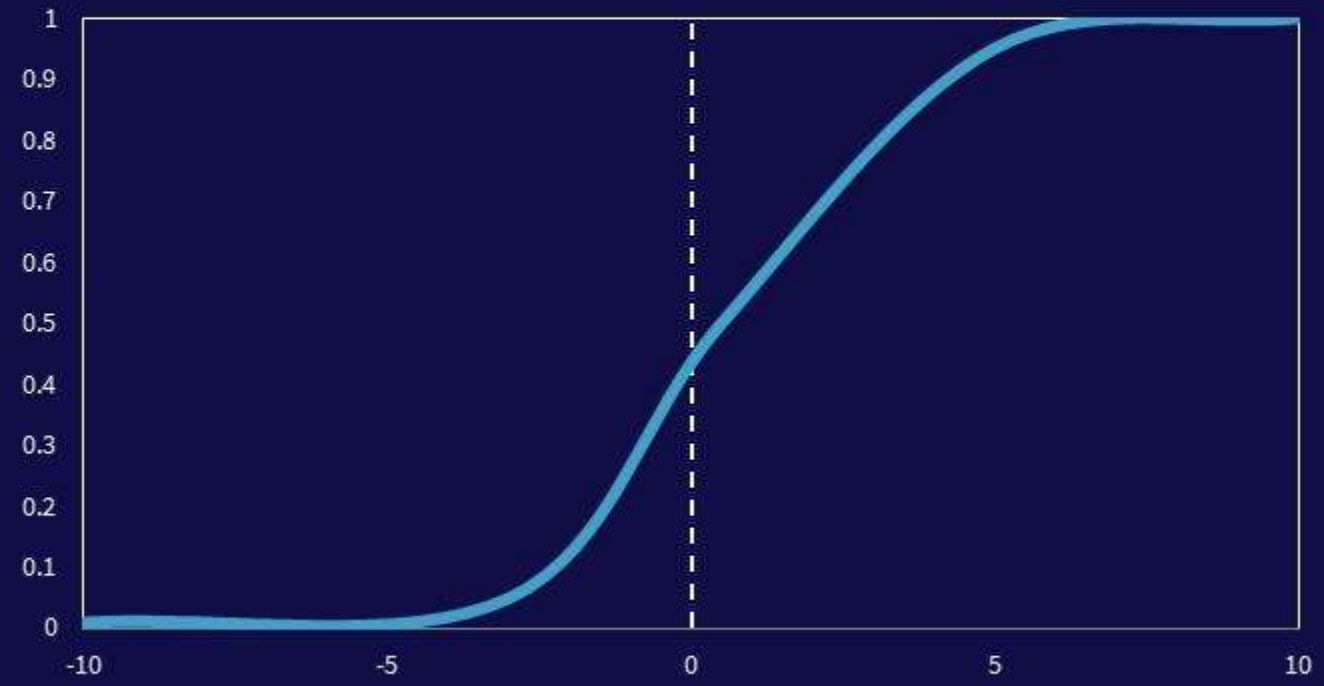


- Most common activation function
- Employed in practically all convolutional neural networks and deep learning functions
- Relu is both derivative as well as monotonic
- Impact the final graph by failing to transfer negative values correctly as each negative input to this activation function is turned into zero in the graph

Sigmoid non-linear activation function

This slide tells about the sigmoid non-linear activation function, which is utilized in models which require the prediction of the possibility of an output.

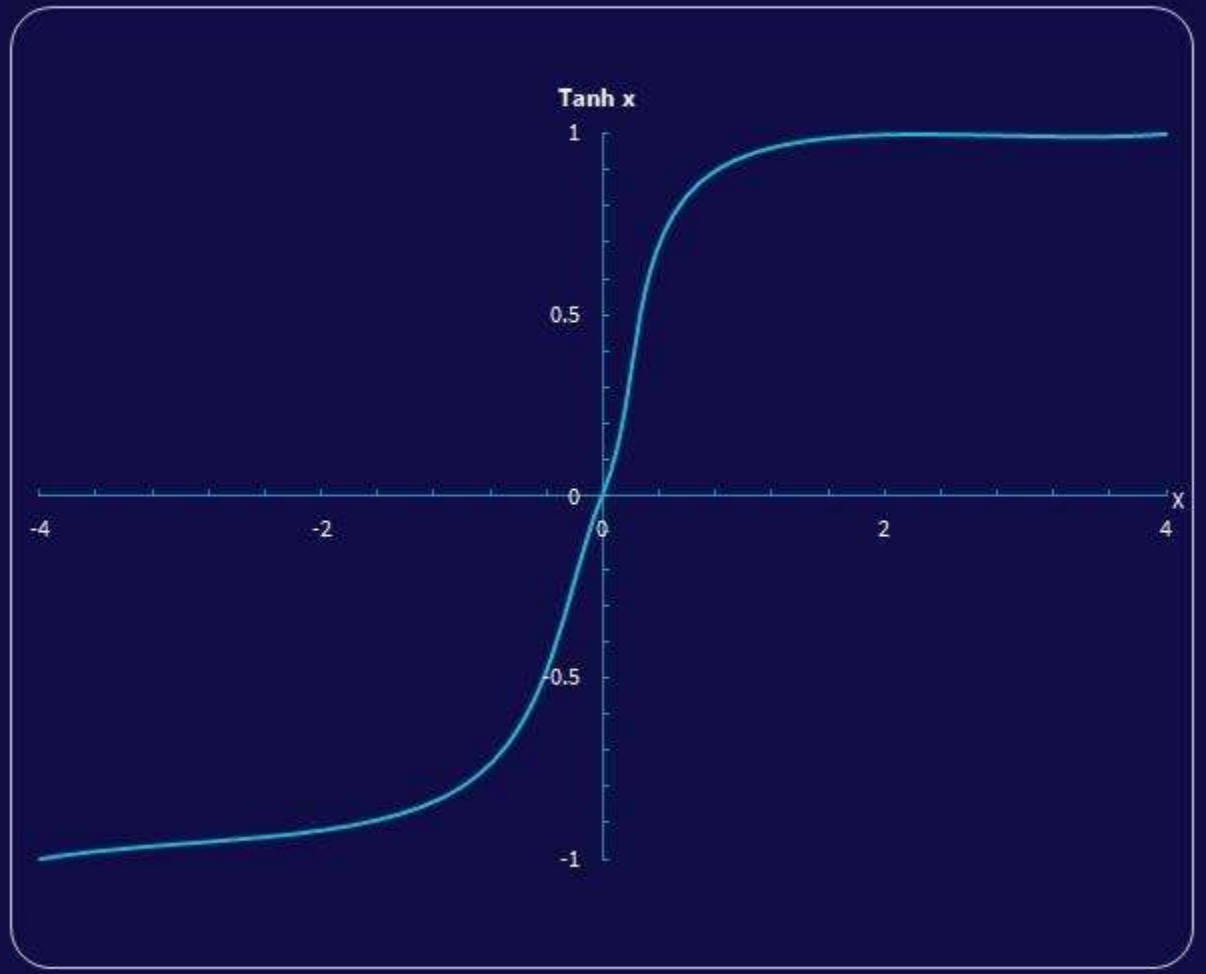
- Utilized in models where the probability must be predicted as an output
- Because the likelihood of anything only occurs within 0 and 1, indeed, sigmoid is the best option
- Has the possibility of causing a neural network to become stuck during training
- Although the function is monotonous, its derivative is indeed not



Hyperbolic tangent(Tanh) activation function

This slide depicts the tanh Activation Function ranging from -1 to 1, and both tanh and sigmoid are used in feedforward networks.

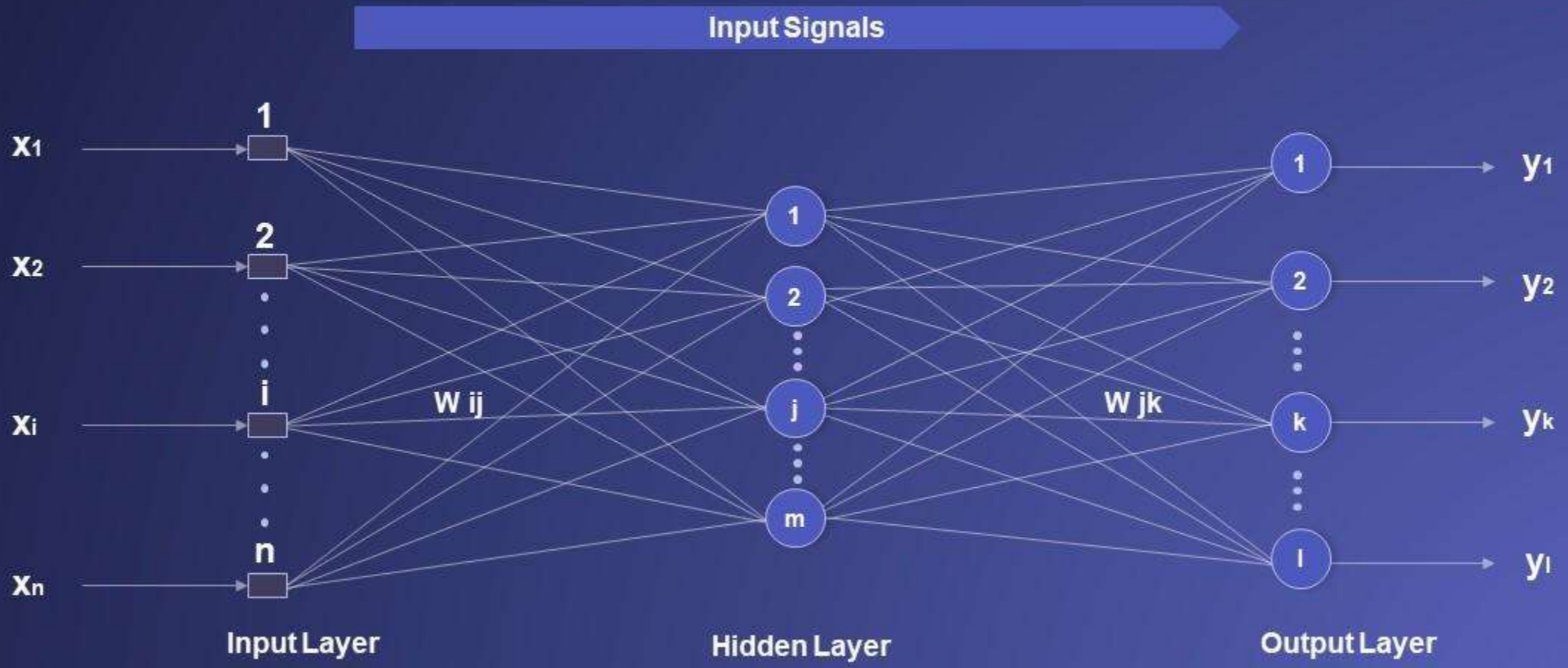
- Tanh function has a range of values between 0 and 1. (-1 to 1) and has a sigmoidal form as well (s-shaped)
- Benefit is that the tanh graph will map the negative inputs firmly negative
- Tanh function is primarily used in classification of two classes



ARTIFICIAL
Neural Networks

Working of artificial neural network

This slide demonstrates the working of Artificial Neural Network, which includes three layers such as input, hidden, and output.



Convolutional Neural Networks



Convolutional Neural Network



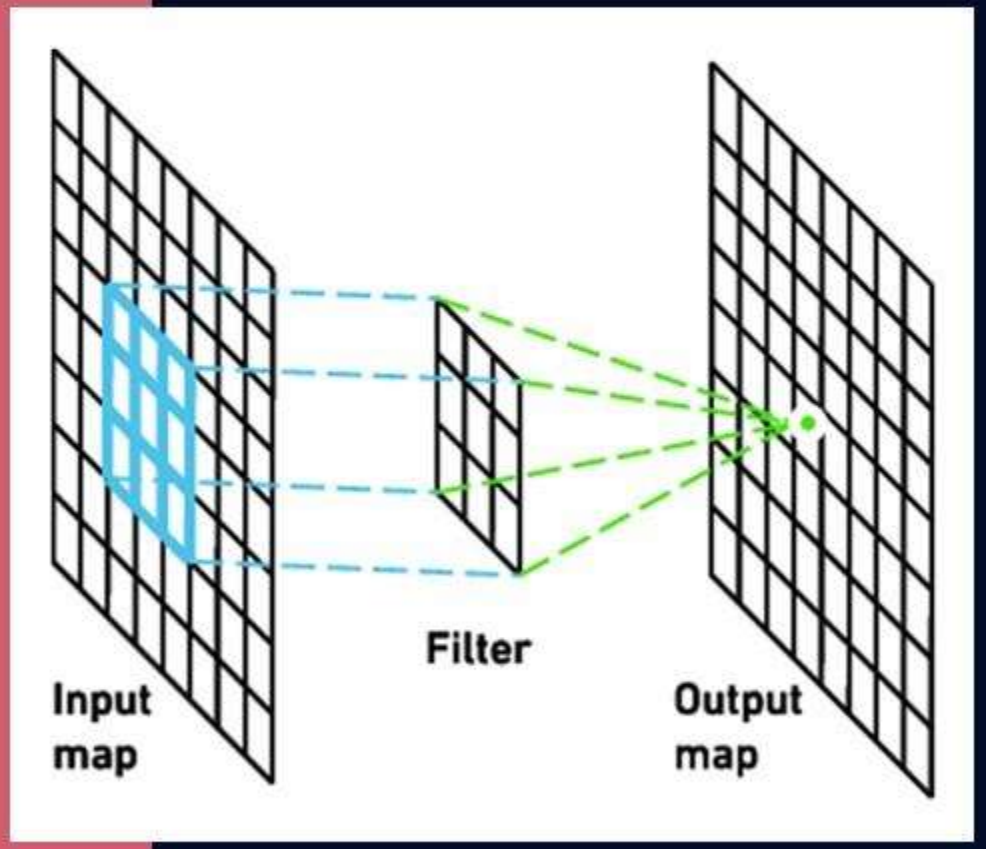
01

Convolutional Neural Network or ConvNet, is a deep learning network design that learns from data without the requirement for human feature extraction

02

CNNs are beneficial for recognizing objects, faces, and settings by looking for patterns in images. They're also helpful for categorizing non-image data, including audio, time series, and signal data

Convolutional Layer in CNN



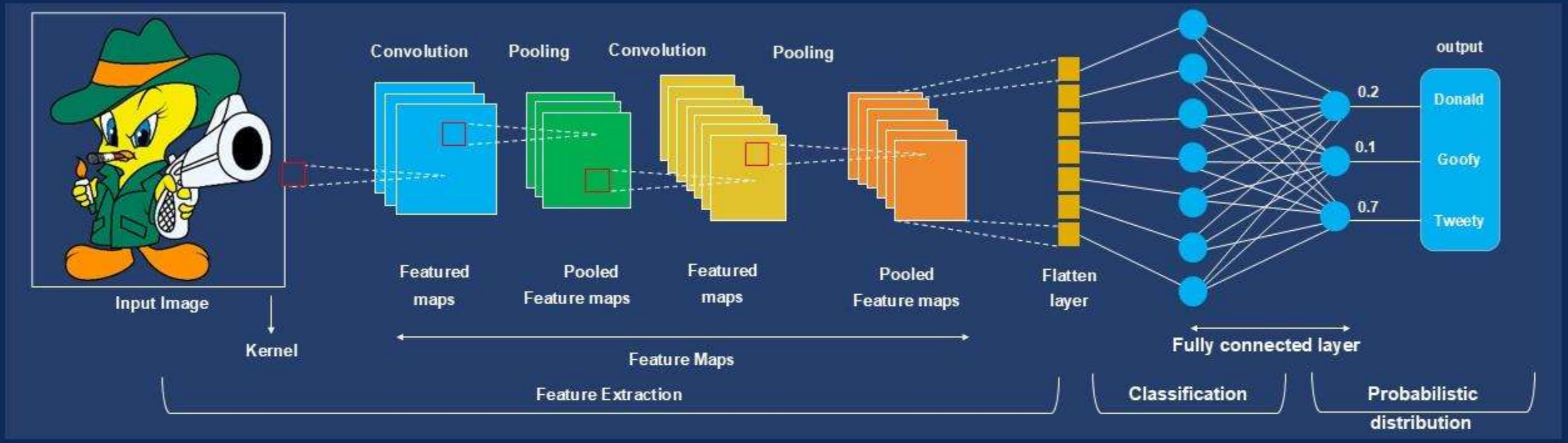
01 Majority of the computation takes place in the convolutional layer of a CNN. This layer requires input data, a filter, and a feature map

02 Consider an input to be a color image composed of a 3D matrix of pixels. This means that the input will have 3 dimensions: width, height, and depth, that match the RGB color space of a picture

03 A feature detector, also termed as a kernel or a filter, scans the image's receptive fields for the presence of a feature. This is known as convolution

How does convolutional neural network works

This slide showcases how convolutional neural networks work which can be referred by IT professionals seeking back-end technology. It provides details about kernel, convolution, pooling, featured maps, output, etc.



01 Convolutional layer

- Uses filters to extract features from input data such as edges, corners or textures
- Helps network understand different patterns and shapes in data

02 Pooling layer

- Reduces size of features obtained from previous layer
- Helps summarize important information and make network focus on most relevant features

03 Fully connected layer

- Takes important features learned from previous layers and makes predictions
- Helps network understand relationship between different features and classify data accurately